



NSE Stock Market Analysis in SQL: Insights Report

Moving averages, golden-cross signals and the data problem hiding in plain sight

Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors

Jan 2015 - Jul 2018 · 889 trading days per stock

Executive summary

- **Scope.** 6 NSE stocks, 889 trading days each, analysed entirely with SQL (24 queries, window functions, CTEs). On bonus-adjusted prices, the 20/50-day golden-cross rule produced **57 Buy and 57 Sell** signals. On raw prices, Task 10 records 56 Buy and 57 Sell signals.
- **The data trap changes the story.** TCS (31 May 2018) and Infosys (15 Jun 2015) each show a one-day fall of about 50%. These are 1:1 bonus issues, not crashes. On raw prices TCS is -23.8% and Infosys -30.9%; after adjustment they are **+52.4%** and **+38.2%**. Winners and losers flip.
- **Best / weakest after adjustment:** TVS Motors (+86.9%) / Hero Motocorp (+6.0%).
- **The signal rule did not beat buy-and-hold** for any stock tested (see section 4). Every one of the 11 trades held under 30 days lost money (average -5.0%).
- **The bonus cliff also corrupted signals.** TCS's latest signal on raw prices was a Sell (5 Jun 2018). On clean prices it is a **Buy (20 Apr 2018)**, and Infosys gains 2 signals.

How to read this report

Every claim in section 3 is given three ways: the **claim**, the **evidence** (which SQL task and which number) and the **caveat** (what could make it wrong). Task numbers refer to `stock_analysis.sql`. Prices are in rupees. Nothing here is investment advice.

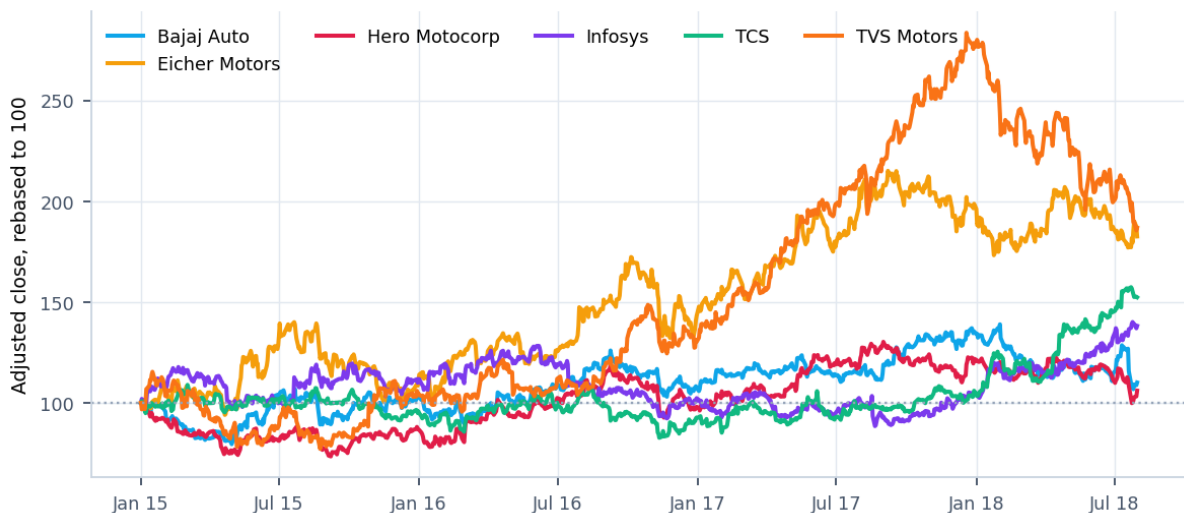
1. Data and method

Each table has one row per trading day (889 rows, 1 Jan 2015 to 31 Jul 2018) with OHLC prices, VWAP, volume, turnover and delivery data. I load the CSVs, convert dates like 31-July-2018 to ISO 2018-07-31 (so text sorting equals time sorting) and run the analysis in SQL:

- **Tasks 1-4** profile the data: history length, Eicher's best closes (all in September 2017), TCS yearly averages and the NULL deliverable_qty rows. These are one row per stock on only two dates (2015-12-09 for four stocks, 2017-08-31 for Infosys), which points to an exchange reporting gap, not company-specific events.
- **Tasks 5-10** build the 20- and 50-day moving averages with `AVG() OVER (... ROWS BETWEEN n PRECEDING)`, blank the incomplete windows, and detect crossings with `LAG()`. A cross is a Buy on the day MA20 moves above MA50 (yesterday it was not), a Sell for the mirror image, otherwise Hold. Task 10 does all stocks in one query using `PARTITION BY`.
- **Tasks 11-13** compute returns, find the data trap (worst one-day move) and fix it by dividing pre-event prices by 2.
- **Tasks 14-24** extend the work: adjusted signals, whipsaw counts, a trade-by-trade backtest, drawdowns, liquidity and yearly returns.

Validation. The signal counts and last-signal dates from my SQL match the course deck exactly for every stock supplied (Eicher 6/7, Hero 9/9, Infosys 9/9, TCS 12/13, TVS 8/8). TCS's 2016 average is exactly 2419.00, TVS is +86.9% and the first signals fall after the 50-day warm-up. So the pipeline reproduces the deck on raw prices, and the differences found later are genuine corrections, not bugs.

Performance at a glance (adjusted prices)



Stock	First close	Last close	Total return	CAGR	Volatility	Max drawdown
Bajaj Auto	₹2,454.10	₹2,700.70	+10.0%	+2.7%	23.1%	-23.2%
Eicher Motors	₹15,239.15	₹27,820.95	+82.6%	+18.3%	28.9%	-28.3%
Hero Motocorp	₹3,107.30	₹3,293.80	+6.0%	+1.6%	22.1%	-26.9%
Infosys	₹987.90	₹1,365.00	+38.2%	+9.5%	23.6%	-31.1%
TCS	₹1,274.10	₹1,941.25	+52.4%	+12.5%	21.4%	-24.1%
TVS Motors	₹276.85	₹517.45	+86.9%	+19.1%	30.6%	-34.6%

2. The data trap: two crashes that never happened

Task 12 ranks each stock's worst day-over-day move. Most stocks' worst day is between -6.2% and -9.7%, but **TCS (-50.4% on 2018-05-31)** and **Infosys (-49.9% on 2015-06-15)** are in a different league. A large profitable company does not lose half its value in a day with no headlines. Searching those names and dates shows both were **1:1 bonus issues**: every holder received one extra share per share held, the share count doubled and the price halved, with no change in anyone's wealth.



How I handled it (task 13)

I divided every close **before** the event date by 2 (TCS before 2018-05-31, Infosys before 2015-06-15) and left later prices unchanged, then rebuilt the moving averages and signals on the adjusted series (tasks 14-15). The cliff disappears and the line is continuous. All other stocks are unchanged.

Stock	Raw return	Adjusted return	Raw signals (B/S)	Adjusted signals (B/S)
TCS	-23.8%	+52.4%	12 / 13	12 / 12
Infosys	-30.9%	+38.2%	9 / 9	10 / 10

Which signals were fake? (task 17)

- Infosys, 2015-07-01: raw says **Hold**, adjusted says **Buy**.
- Infosys, 2015-07-08: raw says **Hold**, adjusted says **Sell**.
- Infosys, 2015-07-27: raw says **Hold**, adjusted says **Buy**.
- Infosys, 2015-08-18: raw says **Buy**, adjusted says **Hold**.
- TCS, 2018-06-05: raw says **Sell**, adjusted says **Hold**.

Why signals far from the event date change. A moving average smears a price cliff across its whole window (20 or 50 days). The TCS Sell on 5 June 2018 came five days after the bonus issue, when the 20-day average had been dragged down by the halved prices. It was an arithmetic artefact. Infosys's raw series also hid real crossings in July and August 2015 for the same reason.

3. Stock by stock: claim, evidence, caveat

■ Bajaj Auto



	Claim	Evidence	Caveat
Price change	Bajaj Auto rose +10.0% (+2.7% a year) over the period.	Task 18: ₹2,454.10 to ₹2,700.70. Max drawdown -23.2% (trough 2018-07-26).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data.
Signals	12 Buy and 11 Sell signals; the latest was a Buy on 2018-06-21.	Tasks 10 and 15. Current MA20 vs MA50: uptrend.	No signal can appear before day 50 (March 2015). Counts match the deck's raw figures.
Whipsaws	11 of 22 consecutive signals came less than 30 days apart (50%). Those 5 quick trades averaged -4.1%.	Tasks 19-20. Shortest gap 5 days. Trades held 30+ days: 6 averaging +4.2%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 11 trades.
Signal vs trend	The long-run move is small (+10.0% in 3.6 years), so there is no clear trend to compare with. The latest signal (Buy, 2018-06-21) is the only directional information, and it is short-term.	Task 10 plus task 18. Current MA20 is above MA50.	A near-flat stock with large swings is exactly where moving-average crossovers whipsaw; read the signal with caution.

■ Eicher Motors



	Claim	Evidence	Caveat
Price change	Eicher Motors rose +82.6% (+18.3% a year) over the period.	Task 18: ₹15,239.15 to ₹27,820.95. Max drawdown -28.3% (trough 2015-11-16).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data.
Signals	6 Buy and 7 Sell signals; the latest was a Sell on 2018-06-06.	Tasks 10 and 15. Current MA20 vs MA50: downtrend.	No signal can appear before day 50 (March 2015). Counts match the deck's raw figures.
Whipsaws	1 of 12 consecutive signals came less than 30 days apart (8%). No completed trade was held under 30 days.	Tasks 19-20. Shortest gap 18 days. Trades held 30+ days: 6 averaging +10.2%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 6 trades.
Signal vs trend	The latest signal (Sell, 2018-06-06) disagrees with the long-run trend (+82.6%). I would treat the 3-year trend as the better guide to direction and the signal as a short-term warning only.	Task 10 plus task 18. Current MA20 is below MA50.	A moving-average cross reacts late; the trend could also resume or the Sell may be an early sign of a genuine reversal.

■ Hero Motocorp



	Claim	Evidence	Caveat
Price change	Hero Motocorp rose +6.0% (+1.6% a year) over the period.	Task 18: ₹3,107.30 to ₹3,293.80. Max drawdown -26.9% (trough 2015-09-07).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data.
Signals	9 Buy and 9 Sell signals; the latest was a Sell on 2018-05-22.	Tasks 10 and 15. Current MA20 vs MA50: downtrend.	No signal can appear before day 50 (March 2015). Counts match the deck's raw figures.
Whipsaws	4 of 17 consecutive signals came less than 30 days apart (24%). Those 1 quick trades averaged -6.5%.	Tasks 19-20. Shortest gap 8 days. Trades held 30+ days: 8 averaging -1.1%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 9 trades.
Signal vs trend	The long-run move is small (+6.0% in 3.6 years), so there is no clear trend to compare with. The latest signal (Sell, 2018-05-22) is the only directional information, and it is short-term.	Task 10 plus task 18. Current MA20 is below MA50.	A near-flat stock with large swings is exactly where moving-average crossovers whipsaw; read the signal with caution.

Infosys



	Claim	Evidence	Caveat
Price change	Infosys rose +38.2% (+9.5% a year) over the period.	Task 18: ₹987.90 to ₹1,365.00. Max drawdown -31.1% (trough 2017-08-21).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data. On raw prices this would wrongly read -30.9%.
Signals	10 Buy and 10 Sell signals; the latest was a Buy on 2018-05-07.	Tasks 10 and 15. Current MA20 vs MA50: uptrend.	No signal can appear before day 50 (March 2015). Adjusted counts differ from raw (9/9 raw).
Whipsaws	6 of 19 consecutive signals came less than 30 days apart (32%). Those 2 quick trades averaged -3.3%.	Tasks 19-20. Shortest gap 3 days. Trades held 30+ days: 7 averaging +0.5%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 9 trades.
Signal vs trend	The latest signal (Buy) agrees with the long-run direction (+38.2%).	Task 10 plus task 18.	Agreement between a 3-year trend and a 2-month signal is weak confirmation, not proof.

TCS



	Claim	Evidence	Caveat
Price change	TCS rose +52.4% (+12.5% a year) over the period.	Task 18: ₹1,274.10 to ₹1,941.25. Max drawdown -24.1% (trough 2016-11-11).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data. On raw prices this would wrongly read -23.8%.
Signals	12 Buy and 12 Sell signals; the latest was a Buy on 2018-04-20.	Tasks 10 and 15. Current MA20 vs MA50: uptrend.	No signal can appear before day 50 (March 2015). Adjusted counts differ from raw (12/13 raw).
Whipsaws	7 of 23 consecutive signals came less than 30 days apart (30%). Those 1 quick trades averaged -4.4%.	Tasks 19-20. Shortest gap 1 days. Trades held 30+ days: 10 averaging -3.3%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 11 trades.
Signal vs trend	The latest signal (Buy) agrees with the long-run direction (+52.4%).	Task 10 plus task 18.	Agreement between a 3-year trend and a 2-month signal is weak confirmation, not proof.

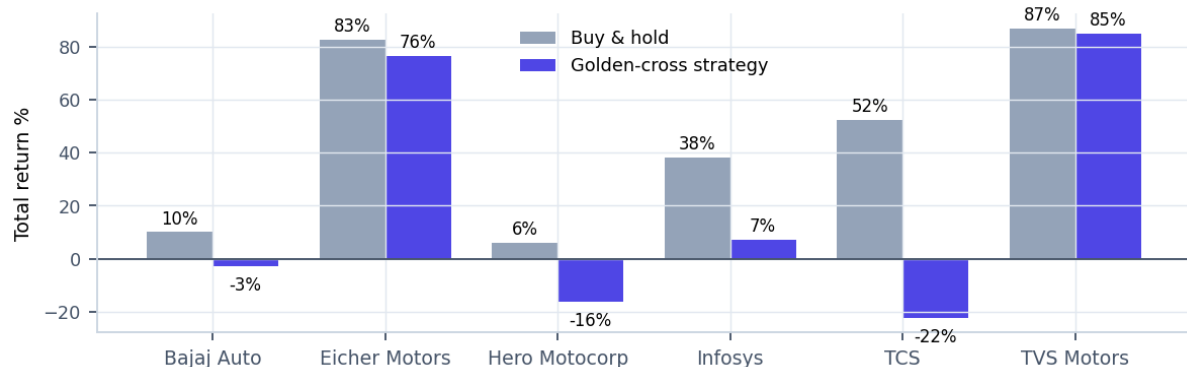
TVS Motors



	Claim	Evidence	Caveat
Price change	TVS Motors rose +86.9% (+19.1% a year) over the period.	Task 18: ₹276.85 to ₹517.45. Max drawdown -34.6% (trough 2018-07-30).	Start and end dates are arbitrary points; dividends and splits other than the two found are not in the data.
Signals	8 Buy and 8 Sell signals; the latest was a Sell on 2018-05-17.	Tasks 10 and 15. Current MA20 vs MA50: downtrend.	No signal can appear before day 50 (March 2015). Counts match the deck's raw figures.
Whipsaws	4 of 15 consecutive signals came less than 30 days apart (27%). Those 2 quick trades averaged -8.3%.	Tasks 19-20. Shortest gap 6 days. Trades held 30+ days: 6 averaging +17.7%.	Returns exclude brokerage, taxes and slippage, which would hurt frequent trading more. Sample is only 8 trades.
Signal vs trend	The latest signal (Sell, 2018-05-17) disagrees with the long-run trend (+86.9%). I would treat the 3-year trend as the better guide to direction and the signal as a short-term warning only.	Task 10 plus task 18. Current MA20 is below MA50.	A moving-average cross reacts late; the trend could also resume or the Sell may be an early sign of a genuine reversal.

4. Would following the signals have paid?

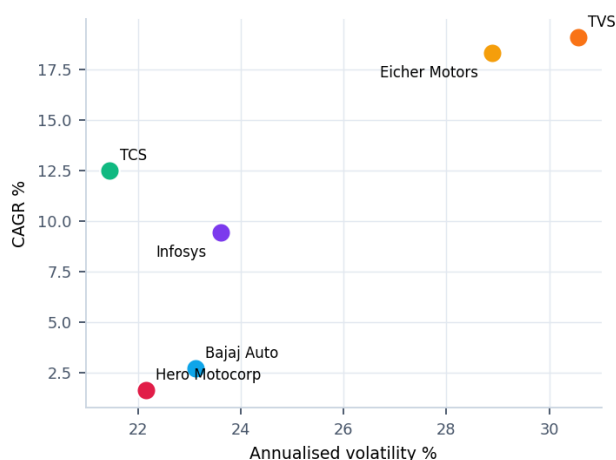
Task 20 builds every completed trade (buy at a Buy signal's close, sell at the next Sell's close). Section 4 uses a stricter daily version: the strategy owns the stock after a Buy and holds cash after a Sell, acting on the next day's return so it never uses information it would not have had.



Stock	Buy & hold	Strategy	Time invested	Trades	Win rate	Avg trade
Bajaj Auto	+10.0%	-2.7%	56%	11	36%	+0.4%
Eicher Motors	+82.6%	+76.5%	52%	6	100%	+10.2%
Hero Motocorp	+6.0%	-16.3%	53%	9	33%	-1.7%
Infosys	+38.2%	+7.1%	53%	9	33%	-0.4%
TCS	+52.4%	-22.4%	50%	11	18%	-3.4%
TVS Motors	+86.9%	+85.0%	57%	8	38%	+11.2%

- The rule beat buy-and-hold for **0 of 6** stocks. It lagged most for TCS (-22.4% vs +52.4%): only 2 of its 11 trades won, in a market that moved sideways until 2017 and then rallied while the strategy was often in cash.
- Where it worked, it worked through **long holds**: Eicher won all 6 of its trades and TVS's 30+ day trades averaged +17.7%.
- Every trade shorter than 30 days lost money (11 of 11), a clear signature of a lagging indicator whipsawing in a range-bound market.

Risk, return and co-movement



Bajaj	1.00	0.33	0.52	0.16	0.16	0.36
Eicher	0.33	1.00	0.28	0.11	0.10	0.28
Hero	0.52	0.28	1.00	0.18	0.18	0.33
Infosys	0.16	0.11	0.18	1.00	0.41	0.13
TCS	0.16	0.10	0.18	0.41	1.00	0.11
TVS	0.36	0.28	0.33	0.13	0.11	1.00
Bajaj	Eicher	Hero	Infosys	TCS	TVS	

Return per unit of risk is highest for **Eicher Motors** (0.63) and lowest for **Hero Motocorp** (0.07). Correlations of daily returns are modest (highest pair 0.52), so the stocks give some diversification.

5. Questions about the method

A moving average only uses past prices. What does that mean for how early a golden cross can tell you anything?

It tells you late, by construction. A 50-day average needs 50 days of history, so no signal is possible before mid-March 2015 for any stock (first real signals appear in late March to June 2015). More importantly, the crossing happens only after the price has already moved. Across all Buy signals here, the price was on average **10.8% above its lowest close of the preceding 60 trading days** by the time the Buy fired: a good part of each rally had already gone. The same lag is why quick reversals produce whipsaws.

Which conclusions in the course deck change after task 13?

Deck statement	Verified result	Change
TCS: -23.8% over the period	+52.4% adjusted	Sign flips. The deck used raw prices through a bonus issue.
Infosys: -3%	+38.2% adjusted (raw was -30.9%)	Sign flips. The deck's -3% is also a typo for -30.9% raw.
TCS start price 2454.1	₹2,548.20 raw / ₹1,274.10 adjusted	The deck repeats Bajaj's start price; it is not TCS's.
TCS trend 'shifting down' (5 Jun 2018)	Latest signal Buy, 20 Apr 2018; MA20 above MA50	The Sell was created by the bonus cliff.
TCS: 12 Buy / 13 Sell	12 Buy / 12 Sell	One fake Sell removed.
Infosys: 9 Buy / 9 Sell	10 Buy / 10 Sell	Cliff hid genuine crossings in mid-2015.
Eicher: +82.5%	+82.6%	Rounding only (82.57 rounds to 82.6).
TVS +86.9%, Hero +6%	Unchanged	Both confirmed.
Final report: sell TCS, favour Infosys	TCS is #3 of 6 by return (+52.4%) and its latest signal is Buy	Recommendation reverses for TCS.

Evidence of a shared market event: on 11 Nov 2016, **4 of 6** stocks fell by more than 3% in a single day, and TCS's drawdown trough and Hero's second-worst day both fall on that date. Price data alone cannot say why, which is exactly the kind of context this dataset lacks.

What isn't in this data that a real investor would need?

- **Dividends:** total return would be higher, especially for Hero, TCS and Infosys which pay regular dividends.
- **Costs:** brokerage, STT, stamp duty, taxes and bid-ask slippage. They punish the whipsaw trades most.
- **News and fundamentals:** earnings, guidance, regulation and corporate actions. The two bonus issues were found only by inspecting the numbers; other splits or demergers could hide elsewhere.
- **The broader market:** no Nifty or sector index, so stock-specific moves cannot be separated from market moves; no risk-free rate for proper risk-adjusted metrics.
- **Regime diversity:** 3.6 years of one mostly rising market. A rule that lags may behave very differently in a long bear or a strongly trending year.

6. Reproducing this work

sql/stock_analysis.sql (SQLite, re-runnable top to bottom) and sql/stock_analysis_mysql.sql (MySQL 8 with the CSV loader and the task 9 function). streamlit run app.py opens the interactive dashboard and live SQL playground running these same queries. Limitations: six stocks over 3.6 years; past performance; not investment advice.

5. Machine Learning Analysis

A separate ML experiment was added to complement the SQL/golden-cross analysis. It predicts next-trading-day direction using only current and historical features, with chronological splitting to avoid future leakage.

Methodology

Feature engineering: 1-day, 5-day and 20-day returns; moving-average ratios; 20-day volatility; high-low and close-open ranges; volume/turnover changes; RSI(14); and delivery percentage. Bonus adjustment is applied before the model dataset is built.

Data split: 3,638 training rows (2015-01-30 to 2017-07-13), 778 validation rows (2017-07-14 to 2018-01-18), and 786 test rows (2018-01-19 to 2018-07-30). TimeSeriesSplit is used for cross-validation and hyperparameters are tuned on the training period.

Model comparison

Model	CV ROC-AUC	Val ROC-AUC	Test Accuracy	Test F1	Test ROC-AUC
Majority Class Baseline	-	-	0.515	0.000	0.500
Logistic Regression	0.528	0.537	0.520	0.445	0.521
HistGradientBoosting	0.512	0.522	0.531	0.495	0.522
Random Forest	0.515	0.520	0.518	0.516	0.516

Selected model and diagnosis

Logistic Regression was selected. Test accuracy was 0.520 and test ROC-AUC was 0.521, only slightly above the majority-class baseline accuracy of 0.515. Train ROC-AUC was 0.539 versus validation 0.537 and test 0.521. The very small train-validation gap and weak overall ROC-AUC indicate potential underfitting rather than a strong predictive model.

Hyperparameter tuning

Logistic Regression: C = 0.01. Random Forest: max_depth = 3, max_features = sqrt, min_samples_leaf = 3, n_estimators = 250. HistGradientBoosting: l2_regularization = 1, learning_rate = 0.05, max_iter = 100, max_leaf_nodes = 15.

ML Appendix — Feature Importance & Interpretation

The selected/diagnostic outputs are stored in the `ml_results/` folder so the experiment is reproducible and auditable.

Feature importance

Feature	Importance
sma_20_ratio	0.1040
ret_1	0.0728
sma_5_ratio	0.0574
delivery_pct	0.0486
vol_20	0.0317
hl_pct	0.0160
ret_5	0.0134
rsi_14	0.0105
volume_chg	0.0095
turnover_chg	0.0071
co_pct	0.0058
ret_20	0.0020

Interpretation and limitation

The strongest engineered feature in this run was `sma_20_ratio` (0.1040), followed by `ret_1` (0.0728) and `sma_5_ratio` (0.0574). These importance values are descriptive, not proof of causality. The test ROC-AUC remains close to 0.50, so the model should not be presented as a reliable trading system. The dataset is a historical sample of six stocks, and the experiment does not model dividends, brokerage, taxes, slippage, news, macro factors, or regime changes.

Submission note

The primary analysis remains SQL-based as required by the student guide. The ML appendix is an additional analytical layer and does not replace the 24 SQL tasks, the bonus-issue adjustment, or the golden-cross backtest.